

ENERGICAL DATA INTELLIGENCE

Customer Segmentation & Revenue Optimization Strategy

Challenge 2

This report explains the reasoning behind the five notebooks that form the ENERGICAL data science workflow.

The data: 17,263 transactions | 4,112 customers | 3 years (2022-2024)

Technical tools: Python, pandas, scikit-learn, statsmodels, Plotly, Streamlit

Models: K-Means clustering, Logistic Regression, Random Forest feature importance, time-series forecasting

APIs: None

Code: github.com/serene-b/energical-datasci

Data Cleaning & EDA

First Notebook establishes the foundation for the entire analytics pipeline by preparing the raw transactional data for further analysis. In any data science or analytics project, the quality of the final models depends directly on the quality of the input data.

The data cleaning process begins by filtering the dataset to retain only completed ("Terminée") orders while excluding returned, pending, and partially refunded transactions. This ensures that all revenue calculations and customer analyses are based solely on finalized sales. Essential variables such as order dates, quantities, and transaction amounts are converted to appropriate data types, while rows with missing critical information are removed. Missing values in non-essential fields, such as product names and payment methods, are replaced with default labels to preserve transaction records. Geographic data is standardized by correcting inconsistencies in wilaya names and removing invalid locations, ensuring accurate regional analyses.

After cleaning, the transactional data is enriched by merging it with the product catalogue using product-name matching, achieving a high match rate of approximately **96%**. The notebook then performs **exploratory data analysis EDA** to identify important business patterns, including monthly revenue trends, top-selling products, revenue distribution across wilayas, customer type comparisons (B2B versus B2C), product category performance, and regional return rates.

Finally, the cleaned and enriched dataset is exported for use in the following notebooks.

Overall, this notebook establishes a reliable analytical foundation by transforming raw operational data into a structured, high-quality dataset while simultaneously providing valuable business insights that support customer segmentation, forecasting, and strategic decision-making later.

RFM Segmentation

Model Configuration

Algorithm: K-Means Clustering

Number of clusters (k): 3

Features: Recency, Frequency, Monetary (RFM)

Customers analyzed: 4,111

Outlier removal: Customers with Frequency > 100 orders

To better understand customer purchasing behaviour, the cleaned transactional data was analyzed using the **Recency–Frequency–Monetary (RFM)** framework. The three RFM variables were used as input features for the **K-Means clustering algorithm (k = 3)** to group customers with similar purchasing patterns. Before training the model, customers with a purchase frequency greater than 100 orders were treated as outliers and removed to prevent them from disproportionately influencing the clustering results. The final analysis was performed on **4,111 unique customers**.

Segmentation Results

Segment	Clients	%	Avg. Recency	Avg. Frequency	Avg. Monetary
	63	1.5%	100 jours	43 commandes	369 181 DA
	2 277	55.5%	227 jours	3 commandes	27 163 DA
	1 766	43%	758 jours	2 commandes	29 239 DA

Business Interpretation

Champions (63 clients)

These customers purchase **frequently**, have bought recently, and generate exceptionally high revenue. Although they represent only **1.5% of the customer base**, they contribute disproportionately to ENERGICAL's overall turnover and

therefore constitute the company's most valuable customers. Maintaining their loyalty should be a strategic priority through **VIP programs, personalized services, exclusive promotions, and early access to new products**, thereby maximizing customer lifetime value.

● **Occasional Customers (2,277 clients)**

This segment represents the **largest proportion of the customer base (55.5%)**. These customers have previously engaged with the company but their purchasing activity has started to decline, as reflected by their increasing recency. Without timely intervention, many are likely to transition into the Lost segment. Targeted retention initiatives, such as reminder emails, personalized product recommendations, or limited-time discounts can encourage repeat purchases and improve customer retention.

● **Low-Engagement Customers (1,766 clients)**

Customers in this segment have been inactive for an average of **758 days**, indicating a very high probability of being lost. Their prolonged inactivity makes re-engagement considerably more challenging than **Occasional Customers**. A carefully designed **reactivation campaign** may recover a portion of these customers; however, those who remain unresponsive should be gradually excluded from future marketing campaigns in order to optimize advertising expenditures and allocate resources toward higher-value customer segments.

Key Takeaway

The RFM segmentation reveals that ENERGICAL's customer base is highly heterogeneous. Rather than applying identical marketing strategies to every customer, the company **can tailor its approach according to customer value and purchasing behaviour**.

FORECAST & RETENTION ANALYSIS:

The objective of this notebook is to support ENERGICAL's strategic planning by predicting future revenue and customer retention. While the previous notebooks focused on understanding historical customer behaviour through segmentation, this notebook introduces **predictive analytics** by answering two key business questions:

- **How much revenue can be expected over the coming months in each wilaya?**
- **How many existing customers are likely to return?**

Combining these two predictions enables the company to improve inventory planning, allocate marketing resources more efficiently, and make data-driven investment decisions

Revenue Forecasting Methodology

Monthly revenue for each wilaya was aggregated over a three-year period (2022–2024) to create individual time series. Rather than forecasting the national revenue as a whole, separate models were developed for each wilaya to capture regional purchasing behaviour and local market characteristics.

Each time series was first decomposed into three components:

- **Trend**, representing the long-term direction of sales.
- **Seasonality**, representing recurring monthly purchasing patterns.
- **Residuals**, representing unexplained random variation.

This decomposition provides valuable insight into whether observed revenue changes originate from long-term business growth, seasonal demand, or unpredictable external factors.

Stationarity tests were then performed to determine whether each time series exhibited stable statistical properties over time. Stationary series are generally easier to model because their behaviour remains consistent throughout the observation period.

Several forecasting models were evaluated using an **80/20 train-test split**, where 80% of the historical observations were used for training and the remaining 20% for model validation. Model performance was compared using **Root Mean Squared Error (RMSE)** and other forecasting metrics before selecting the best-performing model for each wilaya.

Once the optimal model had been selected, a **95% confidence interval** was generated together with a **three-month forecast**. We decided to keep the forecast at three months

because less would not provide enough information for the sales manager to plan ahead, and more than three months would provide an inaccurate forecast due to the volatility of economic data.

Below is an example of a forecasted graph.

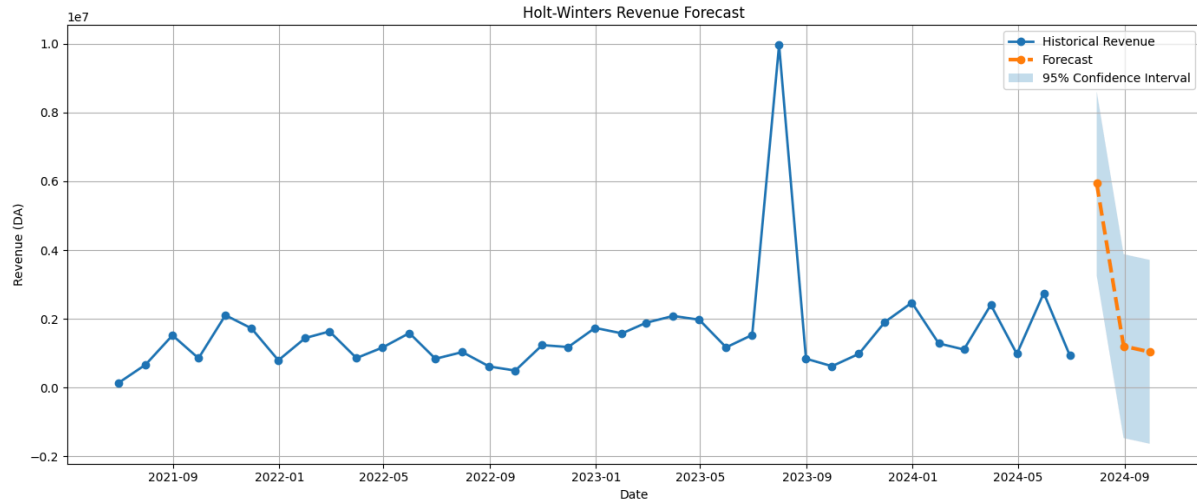
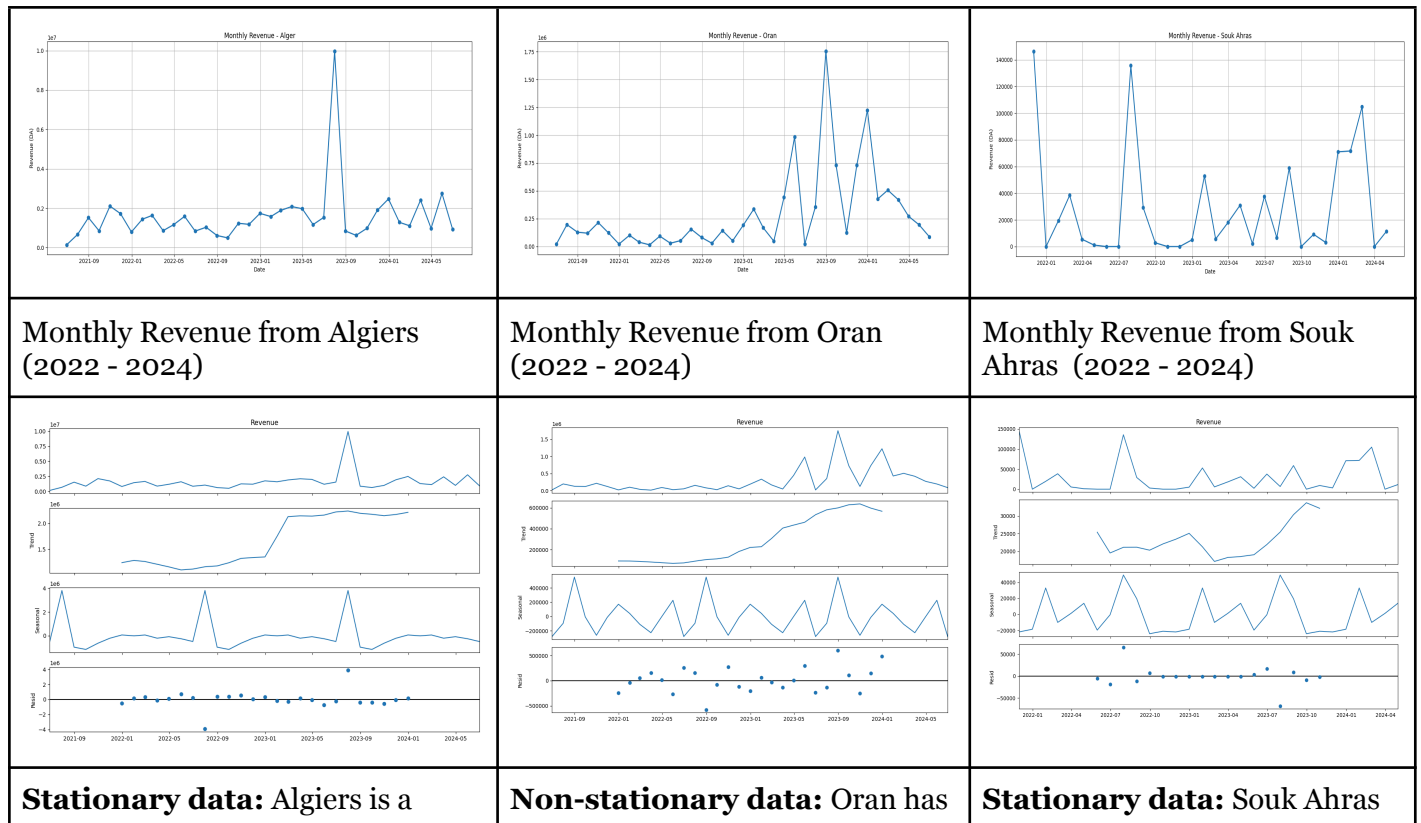


Figure 1: Forecasted data of Algiers' revenue.

Forecast Analysis

Three representative wilayas were selected to illustrate different revenue behaviours.



relatively stable area with predictable increasing trend Clear seasonality (monthly): significant periodic peaks are noticed during August of each year, which could be due to increased purchases of summer-season products Gradual upward trend: due to population growth and expansion. Normal residuals: there is no information left after removing seasonality and trend, the data is stationary and could be modeled with ARIMA.	been relatively stagnant in terms of urban development and expansion until recently (2023) Clear seasonality (monthly): Seasonality is due to buying heaters in the winter Significant upward trend: Due to the housing programs (such as AADL), ongoing urban expansion, private residential renovations, have accelerated. Fluctuations in residuals: There could be information lost due to the unpredictable nature of the data.	is a relatively stable area with predictable constant trend Clear seasonality (monthly): significant periodic peaks are noticed during August of each year, which could be due to increased purchases of summer-season products Constant trend: Due to the low population and therefore low expansion which results in low demand in house-building products Normal residuals: there is no information left after removing seasonality and trend, the data is stationary and could be modeled with ARIMA.
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Table 1:

Top: Examples of the plots of cleaned time series of three wilayas with varying populations, economic activity, and development levels (before training and forecast)

Below: Decomposed time series of each corresponding time series and interpretations of each component.

Forecast Risk Assessment

Revenue alone is insufficient for making business decisions. Two wilayas may generate similar expected revenue while presenting very different levels of uncertainty.

To quantify this uncertainty, the notebook measures **forecast volatility**, defined as:

Mathematically: $\frac{std(\hat{y})}{mean(\hat{y})}$

which translates into the risk taken when making a decision. Intuitively, Lower volatility indicates a stable market with predictable demand, whereas higher volatility reflects greater planning risk. This risk indicator is incorporated into the final recommendation system so that business decisions balance expected revenue against forecasting uncertainty.

Customer Retention Analysis

Objective

The goal of this section is to study customer loyalty and highlight the main factors that affect it, in order to finally predict how many future customers would be returning. The information gained from this analysis and the previous one would immensely improve **inventory maintenance**, since we would be able to predict how much of each product one would need to stock up on before the demand hits.

That way, we avoid overstocking and especially avoid being out of stock, which almost ensures losing even an already loyal customer.

Also, combining this information with the previous analysis would help us understand which wilaya has the most stable customer base, to know where to invest more and deploy more marketing campaigns.

Model Results

The customer retention model was evaluated using **Logistic Regression**, which predicts whether a customer is likely to be **Returning** or **New** based on historical purchasing behaviour. Since customer retention is a binary outcome, Logistic Regression provides a simple and interpretable baseline model for estimating return probabilities.

The model achieved the following performance:

Using this model, **4,112 customers** were analyzed, producing the following predictions:

- **Predicted Returning Customers:** 1,577
- **Predicted Returning Rate:** 38.35%
- **Average Return Probability:** 47.86%

Although the model demonstrates moderate predictive performance, it provides a practical estimate of future customer retention. These predictions can be combined with the revenue forecasts to better anticipate future demand and support inventory planning.

Understanding the Drivers of Customer Loyalty

While Logistic Regression estimates **whether** a customer is likely to return, it does not clearly identify **which factors influence that decision the most**. To address this, a **Random Forest** analysis was performed to identify the variables that contribute most strongly to customer retention.

Rather than focusing solely on prediction, this analysis provides additional business insight by highlighting the characteristics that are most closely associated with returning customers. Understanding these drivers enables ENERGICAL to design more targeted retention strategies and improve customer relationship management by focusing on the factors that have the greatest impact on customer loyalty.

Conclusion

The forecasting and retention analyses complement one another by addressing two essential aspects of business planning. The forecasting model estimates **where and when future demand is expected**, while the retention analysis estimates **how much of that demand is likely to come from returning customers**.

Together, these analyses provide valuable support for inventory management, marketing planning, and strategic decision-making. By anticipating both future sales and customer loyalty, ENERGICAL can optimize stock levels, reduce the risk of shortages or overstocking, and allocate marketing resources more effectively across different wilayas.

Business Recommendations

Based on the EDA, RFM segmentation, revenue forecasting, and retention analysis, ENERGICAL should prioritize the following actions:

1. Protect high-value customers through a VIP retention program

The **Champions** segment represents only **1.5%** of customers but generates the **highest average monetary value**, with an average spending of **369,181 DA** and **43 orders per customer**. These customers should receive priority treatment through exclusive offers, early access to high-demand products, and personalized support.

2. Convert Occasional Customers into repeat buyers

The **Occasional Customers** segment represents **55.5%** of the customer base. This is the largest commercial opportunity because these customers are still active but purchase less frequently than Champions. ENERGICAL should target them with reminder campaigns, personalized recommendations, and limited-time promotions to increase their purchase frequency.

3. Reduce marketing waste on Low-Engagement Customers

The **Low-Engagement Customers** segment represents **43%** of the customer base and has an average recency of **758 days**. A reactivation campaign can be tested, but customers who remain inactive should receive fewer marketing resources so the company can focus its budget on higher-potential segments.

4. Plan inventory according to seasonal demand

The time-series analysis shows clear seasonal patterns in several wilayas. ENERGICAL should increase stock levels before demand peaks, particularly for

product categories linked to winter demand such as heating-related products. This helps avoid stockouts, which can lead to lost sales and weaker customer loyalty.

5. Prioritize campaigns by wilaya using revenue and risk together

Campaigns should not be based on revenue alone. The forecast volatility score should also be considered. Wilayas with high expected revenue and low volatility are safer investment targets, while high-revenue but high-volatility wilayas require more cautious inventory planning.

6. Use a three-month planning cycle

The three-month forecast horizon is appropriate because it gives the sales manager enough time to plan procurement and campaigns while avoiding the unreliability of longer-term forecasts. ENERGICAL should update the dashboard and models every month as new transaction data becomes available.